Week 46: Decision Trees, Bagging and Random Forests

A single tree and the mathematics of CART, the Gini index and the entropy by hand, a tree from scratch – then many trees: why averaging helps, bagging, and the random forest

Morten Hjorth-Jensen

Department of Physics and Center for Computing in Science Education,
University of Oslo, Norway

Plan for week 46

Monday lecture (2×45 min), the last new topic of the semester: Chapter 7, Sections 7.1–7.10.

1. **A single tree**: what it is, regression and classification trees, the three impurity measures and why the Gini index and the entropy are preferred.
2. **The CART algorithm**: the cost it minimises, the gain it maximises – done by hand on ten points, with both criteria.
3. **A worked example**: the ride data – Gini index and entropy at the root and down the tree, and the same tree in `scikit-learn`.
4. **A decision tree from scratch**: forty lines of Python, checked against `scikit-learn`; regression trees; strengths and weaknesses.
5. **Many trees**: why averaging helps (one formula that explains the whole chapter), **bagging** and the out-of-bag error.
6. **Random forests**: the one tweak that decorrelates the trees, own code and `scikit-learn`, two data sets.

Tuesday (flipped classroom): about 20 minutes of questions and answers on today's lecture, then **Project 3**.

Week 47: boosting – AdaBoost, gradient boosting, XGBoost (Sections 7.11–7.15).

A change of plan: the gated recurrent cells (LSTM and GRU) announced for today get a separate lecture later; today is trees and forests only.

Reading, and where the code is

Chapter 7, Sections 7.1–7.10

Basics of a tree (7.1), regression trees and pruning (7.2), classification trees (7.3), the CART algorithm (7.4), the Gini worked example (7.5), implementing a decision tree (7.6), strengths and weaknesses (7.7), why averaging helps (7.8), bagging (7.9), random forests (7.10). Boosting (7.11–7.15) is next week.

Other sources

Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, Chapters 9.2, 8.7 and 15; James et al., *An Introduction to Statistical Learning*, Chapter 8; Géron, *Hands-On Machine Learning*, Chapters 6–7; Raschka et al., Chapter 3 (trees) and 7 (ensembles).

Code

Everything shown today, including the from-scratch tree, bagging and forest, is in `week46.ipynb` (the functions of `make_week46_figures.py`). The book's versions are in `BookPrograms/chapter07_trees_and_ensembles/`.

Every method so far fitted one global function

What we have done

Linear regression: a hyperplane. Logistic regression: a sigmoid of a hyperplane. Neural networks: compositions of such maps. In every case *one* function is adjusted to all the data at once, by a gradient.

What a tree does instead

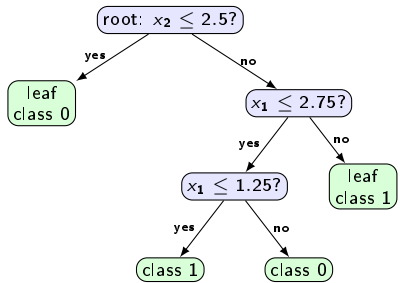
It **partitions** the feature space into boxes by a sequence of yes-or-no questions, and fits a **constant** in each box. No gradient, no scaling, no distance.

Why the chapter is interesting

A tree can be read aloud – the most interpretable model in the course – and, on its own, one of the least accurate: **low bias, very high variance**. Fixing that gives bagging, random forests and boosting: on tabular data, the most accurate methods known.

The vocabulary (Section 7.1)

- The **root node** holds all the observations.
- Each **internal node** tests one feature against a threshold, $x_j \leq t$, and sends the observation left or right along a **branch**.
- Each **leaf** (terminal node) carries a prediction: a class, or a number.
- Predicting a new point = walking from the root to a leaf: one comparison per level, so the cost is logarithmic in the number of leaves.
- The **depth** is the length of the longest walk.



The tree we shall grow by hand in Part 2: depth 3, four leaves.

The geometry: boxes

Each test cuts the space with an axis-aligned plane

$x_j \leq t$ is a hyperplane perpendicular to axis j. A tree therefore partitions the feature space into **axis-aligned boxes**, one per leaf, and predicts a constant in each box.

The limitation

Boxes cannot represent a diagonal boundary economically: a tree approximates $x_1 + x_2 \leq 1$ by a **staircase**, and needs many leaves to make the steps small.

The strength

No notion of distance is needed, so **no scaling** of the features; categorical variables and interactions are handled without any special provision; a monotone transformation of any feature leaves the tree *unchanged* (which no other method in this course can claim).

Regression trees (Section 7.2)

Two steps

(1) Split the predictor space into J non-overlapping boxes $R_1, \ldots, R_J$. (2) In box R_j predict the **mean** $\overline{y}_{R_j}$ of the training responses that fall in it.

The goal: boxes that minimise the squared error, Eq. (7.1)

$$\sum_{j=1}^{J} \sum_{i \in R_j} \left(y_i - \overline{y}_{R_j} \right)^2 .$$

That the best constant in a box is the mean is not an assumption: minimising $\sum_i (y_i - c)^2$ over c gives $c = \overline{y}$ – the one-line calculation that produced the sample mean in Chapter 2.

The problem

The number of ways of partitioning the space into J boxes grows combinatorially. We cannot search them all, so we proceed **top-down** and **greedily**.

Recursive binary splitting, Eqs. (7.2)–(7.3)

One step

Choose a feature x_j and a cutpoint s, splitting the current region into the two half spaces

$$R_1(j,s) = \{X \mid x_j < s\}, \qquad R_2(j,s) = \{X \mid x_j \geq s\},$$

so as to minimise

$$\sum_{i:\, x_i \in R_1} \left(y_i - \overline{y}_{R_1}\right)^2 + \sum_{i:\, x_i \in R_2} \left(y_i - \overline{y}_{R_2}\right)^2,$$

over *all* features $j = 1, \ldots, p$ and *all* cutpoints s.

Only $n-1$ cutpoints per feature matter

The criterion is constant between consecutive sorted values of x_j, so only the midpoints need be tried: one sort per feature, $\mathcal{O}(pn \log n)$ for a split.

Then repeat

Split one of the two regions just made, then one of the three, … until a stopping rule (a minimum number of points per leaf, a maximum depth). **Greedy**: the best split *now*, never looking ahead.

Pause and think 1: what does a tree do between two data points?

Question

A regression tree on one feature x has training points at $x = 1, 2, 3$. Which thresholds t in "$x \leq t$" does the algorithm try, and why not every real number? And what does the fitted tree predict at $x = 10$?

Pause and think 1: what does a tree do between two data points?

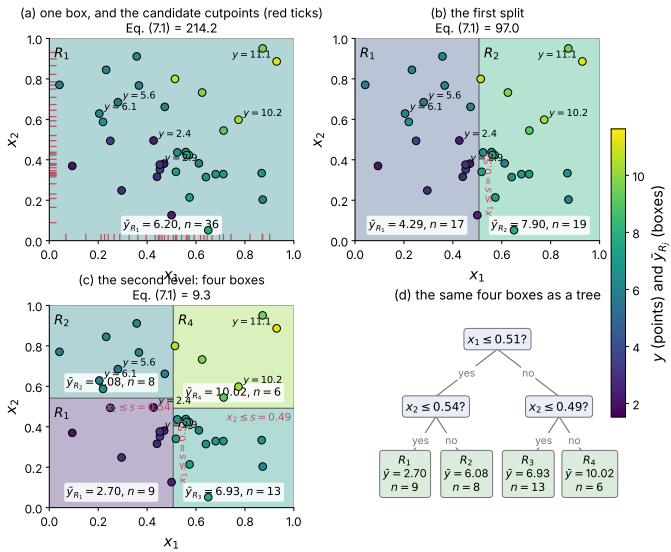
Question

A regression tree on one feature x has training points at $x = 1, 2, 3$. Which thresholds t in "$x \leq t$" does the algorithm try, and why not every real number? And what does the fitted tree predict at $x = 10$?

Answer

Only the midpoints $t = 1.5$ and $t = 2.5$: every $t \in (1, 2)$ puts the same points left and right, so the criterion (7.3) does not change; the midpoint is a convention. At $x = 10$ the point falls into the rightmost leaf and gets that leaf's mean – a tree **cannot extrapolate**; every prediction is an average of observed targets. Keep this in mind for the forests at the end of the lecture: they inherit it.

Recursive binary splitting in pictures (week46_boxes.pdf)



Two features, 36 points.

The response y is the **colour of the points** (a few are labelled); the box mean $\overline{y}_{R_j}$ is the **colour of the box**.

Red ticks: the candidate cutpoints, the midpoints between consecutive values of each feature.

Red labels: the cutpoints chosen.

The number under each title is Eq. (7.1), the sum of squares the tree is minimising:
$214 \to 97 \to 9.3$.

Reading the picture: one split at a time

(a) One box

Everything starts in R_1, predicting the grand mean $\bar{y} = 6.20$; Eq. (7.1) is the total sum of squares, 214. The red ticks are the *candidate* cutpoints: the $n - 1 = 35$ midpoints between consecutive values of x_1 (bottom edge) and of x_2 (left edge) – 70 splits to score, each by Eq. (7.3).

(b) The first split

The best of the seventy is $x_1 \leq 0.51$: two boxes with means 4.29 and 7.90, Eq. (7.1) down to 97. Every point left of the line is now predicted by 4.29, every point right of it by 7.90.

(c) The second level

Each child is split *on its own data*: $x_2 \leq 0.54$ on the left, $x_2 \leq 0.49$ on the right – two different cutpoints for the same feature, because each box only sees its own points. Four boxes, four means 2.70, 6.08, 6.93, 10.02; Eq. (7.1) down to 9.3.

(d) The same thing as a tree

A new point is sent down two questions and receives the mean of the box it lands in. The data were generated as $3 + 4\,[x_1 > 0.5] + 3\,[x_2 > 0.5]$ plus noise of standard deviation 0.5: the tree has found the right boxes and the right plateaus, up to the noise – and would keep splitting the noise if we let it.

Grown to completion, a tree learns nothing: pruning, Eq. (7.4)

The overfitting signature

Let the splitting run and every training point ends in its own leaf: zero training error, and by Chapter 2 no generalisation. A smaller tree has more bias but less variance.

Cost-complexity pruning

Grow a large tree T_0, then for a parameter $\alpha \geq 0$ find the subtree $T \subset T_0$ minimising

$$\sum_{m=1}^{\overline{T}} \sum_{i:\, x_i \in R_m} \left(y_i - \overline{y}_{R_m}\right)^2 \;+\; \alpha\, \overline{T}, \qquad \overline{T} = \text{number of leaves.}$$

This is training error plus α times a complexity penalty: the structure of **Ridge regression**, Eq. (3.43), with the number of leaves in place of $\|\boldsymbol{\theta}\|^2$. Choose α by cross-validation, exactly as in week 36. (`scikit-learn`: `ccp_alpha` and `cost_complexity_pruning_path`.)

In practice today

We shall mostly limit `max_depth` and `min_samples_leaf` instead – and, from Part 5 on, grow trees deep on purpose and let an ensemble deal with the variance.

Classification trees (Section 7.3)

Two differences from a regression tree

A leaf predicts the **majority class** of the training points in it (and the class proportions give a crude probability). And the squared error cannot score a split, so we need a measure of how *impure* a node is.

The class proportions in a node, Eq. (7.5)

For K classes and the N_m points of region R_m,

$$p_{mk} = \frac{1}{N_m} \sum_{x_i \in R_m} I(y_i = k), \qquad k(m) = \text{the majority class.}$$

Three impurity measures, Eqs. (7.6)–(7.8)

$$E_m = 1 - p_{m,k(m)}, \qquad G_m = \sum_{k=1}^{K} p_{mk}(1 - p_{mk}) = 1 - \sum_{k=1}^{K} p_{mk}^2, \qquad S_m = -\sum_{k=1}^{K} p_{mk} \log_2 p_{mk}.$$

Misclassification rate, **Gini index**, **entropy**. All vanish for a pure node; all are maximal when the classes are balanced.

What the Gini index measures

Two readings of $G_m = \sum_k p_{mk}(1 - p_{mk}) = 1 - \sum_k p_{mk}^2$

- **A random guess.** Draw two observations from the node *with replacement*; the probability that they belong to different classes is $1 - \sum_k p_{mk}^2 = G_m$. Equivalently: label a point by drawing a class at random from the node's own proportions – the error rate of that randomized classifier is $\sum_k p_{mk}(1 - p_{mk})$. The majority vote has error $E_m = 1 - \max_k p_{mk}$, always smaller; G_m measures how far the node is from being a safe bet.
- **A variance.** For the indicator $I(y = k)$, a Bernoulli variable with mean p_{mk}, the variance is $p_{mk}(1 - p_{mk})$. So G_m is the *total variance of the one-hot encoded target* inside the node – the exact analogue of the node MSE (7.12) of a regression tree. Splitting to reduce the weighted Gini index is splitting to reduce within-node variance, for classes.

Numbers

Two classes, p the fraction of class 1: $G = 2p(1 - p)$, zero at $p = 0, 1$ and maximal $(1/2)$ at $p = 1/2$. K balanced classes: $G = 1 - 1/K$. Always $0 \leq G \leq 1 - 1/K < 1$. The ride-data root: $1 - (10/14)^2 - (4/14)^2 = 0.408$.

What the entropy measures, and why the two agree

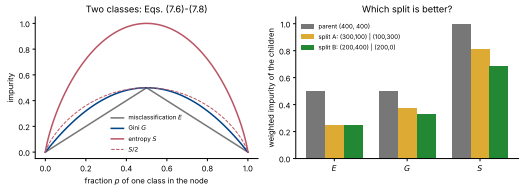
Information: $S_m = -\sum_k p_{mk} \log_2 p_{mk}$

- Shannon: a class with probability p carries $-\log_2 p$ bits of *surprise*; S_m is the **average surprise**, the average number of yes/no questions needed to learn the class of a point drawn from the node. A pure node: 0 bits. Two balanced classes: 1 bit. K balanced classes: $\log_2 K$ bits – unlike G it is not bounded by one.
- The **information gain** $\Delta = S_{\text{parent}} - \sum_c \frac{m_c}{m} S_c$ is the *mutual information* between the split variable and the class: how many bits the answer to the question tells you about the label. The ride data: Outlook answers 0.17 bits of the 0.86 at the root.
- Same object as the cross entropy of Chapter 5: a leaf that predicts its own proportions p_{mk} has cross entropy S_m on its own points, so minimizing the weighted entropy of the children **maximizes the multinomial log-likelihood** of the tree – growing by entropy is maximum likelihood, one split at a time.

Why they rarely disagree

Expand $-p \ln p$ about $p = 1$: $-p \ln p = p(1-p) + \frac{1}{2}p(1-p)^2 + \ldots$, so $S \approx G / \ln 2$ to leading order – the dashed $S/2$ in the figure. Both are strictly concave in p; the error rate $E = 1 - \max_k p_{mk}$ is only piecewise linear, which is the whole difference (next frame).

Why the Gini index and the entropy, not the error rate



Two classes, p = fraction of class 1:
$E = \min(p, 1-p)$ is piecewise linear; $G = 2p(1-p)$ and $S = -p\log_2 p - (1-p)\log_2(1-p)$ are **strictly concave**.

A node with $(400, 400)$ and two candidate splits:
A: $(300, 100) \,|\, (100, 300)$, B: $(200, 400) \,|\, (200, 0)$.

	E	G	S
parent	0.500	0.500	1.000
after A	0.250	0.375	0.811
after B	0.250	**0.333**	**0.689**

Both splits leave the error rate at 0.25: E cannot tell them apart. But B makes a **pure** node, and G and S prefer it. Concavity is the reason: the weighted average of the children is strictly below the parent unless the split is trivial, so a split that purifies one branch is rewarded even when no majority vote changes. A pure node never needs splitting again.

Pause and think 2: the numbers behind the three measures

Question

A node holds three classes with 12, 4 and 4 points. Compute E, G and S. Then: what are the largest possible values of G and S for three balanced classes, and for K balanced classes?

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Answer

$p = (0.6, 0.2, 0.2)$: $E = 1 - 0.6 = 0.4$; $G = 1 - (0.36 + 0.04 + 0.04) = 0.56$;
$S = -(0.6 \log_2 0.6 + 2 \cdot 0.2 \log_2 0.2) = 0.442 + 0.929 = 1.371$ bits.

Balanced, $p_k = 1/K$: $G = 1 - 1/K$ ($= 2/3$ for $K = 3$) and $S = \log_2 K$ ($= 1.585$ for $K = 3$). So $G \leq 1$ always, while S grows with the number of classes – which is why the figure drew $S/2$: for two classes the two curves are nearly proportional, and in practice they rarely choose different splits.

The CART algorithm (Section 7.4)

Classification And Regression Tree – what `scikit-learn` implements

At a node with m points, try every feature k and every threshold t_k, split into left ($x_k \le t_k$) and right, and pick the pair minimising

$$C(k, t_k) = \frac{m_{\text{left}}}{m} G_{\text{left}} + \frac{m_{\text{right}}}{m} G_{\text{right}}, \tag{7.9}$$

the size-weighted impurity of the two children. Then do the same in each child, recursively.

Equivalently, maximise the impurity decrease – the *gain*, Eq. (7.10)

$$\Delta = G_{\text{parent}} - \frac{m_{\text{left}}}{m} G_{\text{left}} - \frac{m_{\text{right}}}{m} G_{\text{right}}.$$

With the entropy in place of G, Δ is the **information gain**, and choosing splits by it (one branch per value of a categorical feature) is the **ID3** algorithm. Same idea, different impurity, multiway splits.

Stopping

A maximum depth, a minimum number of points to split a node or to make a leaf, a maximum number of leaves, or no split with $\Delta > 0$.

CART for regression, Eqs. (7.11)–(7.12)

Replace the impurity by the mean squared error of the node

$$C(k, t_k) = \frac{m_{\text{left}}}{m} \operatorname{MSE}_{\text{left}} + \frac{m_{\text{right}}}{m} \operatorname{MSE}_{\text{right}}, \qquad \operatorname{MSE}_{\text{node}} = \frac{1}{m_{\text{node}}} \sum_{i \in \text{node}} \left(y_i - \overline{y}_{\text{node}}\right)^2 .$$

This is exactly the split criterion (7.3) divided by m, so the two formulations agree.

One algorithm, three impurities

task	node impurity	leaf prediction
regression	$\operatorname{MSE}_{\text{node}}$ (7.12)	mean of y
classification	Gini G (7.7) or entropy S (7.8)	majority class

Everything else is bookkeeping

Sort each feature once, scan the midpoints, keep the best (k, t_k), recurse. That is what we shall write in Part 4.

The algorithm as a recipe

grow(node with data (X, y), depth d)

1. Compute the node's impurity G_{parent} and its leaf prediction (majority class or mean).
2. If $d = d_{\max}$, or too few points, or the node is pure: return a leaf.
3. For every feature k: sort its values, and for every midpoint t between consecutive distinct values compute $C(k, t)$ of Eq. (7.9) and the gain $\Delta = G_{\text{parent}} - C(k, t)$.
4. Keep the (k, t) with the largest Δ. If $\Delta \leq 0$: return a leaf.
5. Split the data by $x_k \leq t$ and call grow on both halves with depth $d + 1$.

Two remarks

Ties happen (two splits with the same gain); CART breaks them by order or at random, and that is one reason two implementations can grow different trees from the same data. And the search is **greedy**: the globally best tree is NP-complete to find; nobody tries.

CART by hand: ten points in the plane

point	x_1	x_2	class
1	1.0	1.0	0
2	2.0	2.0	0
3	3.0	2.0	0
4	1.5	4.0	0
5	5.0	1.0	0
6	4.0	4.0	1
7	5.0	3.0	1
8	6.0	5.0	1
9	1.0	5.0	1
10	5.5	4.5	1

The root

Five of each class: $p = (0.5, 0.5)$, so $G_{\text{root}} = 0.5$ and $S_{\text{root}} = 1$ bit.

The candidates

x_1 takes the distinct values $1, 1.5, 2, 3, 4, 5, 5.5, 6$: seven midpoints $1.25, 1.75, 2.5, 3.5, 4.5, 5.25, 5.75$. x_2 takes $1, 2, 3, 4, 4.5, 5$: five midpoints $1.5, 2.5, 3.5, 4.25, 4.75$. **Twelve** splits to score.

One of them, worked out

$x_2 \le 2.5$: left = points $1, 2, 3, 5$, all class 0, $G_{\text{left}} = 0$;
right = points $4, 6, 7, 8, 9, 10$, one class 0 and five class 1,
$G_{\text{right}} = 1 - (1/6)^2 - (5/6)^2 = 10/36 = 0.2778$.
$C = \frac{4}{10} \cdot 0 + \frac{6}{10} \cdot 0.2778 = 0.1667$,
$\Delta = 0.5 - 0.1667 = 0.3333$.

All twelve candidates at the root

split	n_{L}	n_{R}	C_{Gini}	Δ_{Gini}	C_{ent}	Δ_{ent}
$x_1 \leqslant 1.25$	2	8	0.5000	0.0000	1.0000	0.0000
$x_1 \leqslant 1.75$	3	7	0.4762	0.0238	0.9651	0.0349
$x_1 \leqslant 2.5$	4	6	0.4167	0.0833	0.8755	0.1245
$x_1 \leqslant 3.5$	5	5	0.3200	0.1800	0.7219	0.2781
$x_1 \leqslant 4.5$	6	4	0.4167	0.0833	0.8755	0.1245
$x_1 \leqslant 5.25$	8	2	0.3750	0.1250	0.7635	0.2365
$x_1 \leqslant 5.75$	9	1	0.4444	0.0556	0.8920	0.1080
$x_2 \leqslant 1.5$	2	8	0.3750	0.1250	0.7635	0.2365
$x_2 \leqslant 2.5$	4	6	**0.1667**	**0.3333**	**0.3900**	**0.6100**
$x_2 \leqslant 3.5$	5	5	0.3200	0.1800	0.7219	0.2781
$x_2 \leqslant 4.25$	7	3	0.2857	0.2143	0.6042	0.3958
$x_2 \leqslant 4.75$	8	2	0.3750	0.1250	0.7635	0.2365

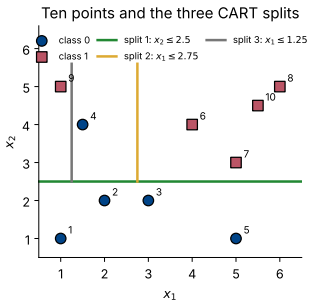
Reading the table

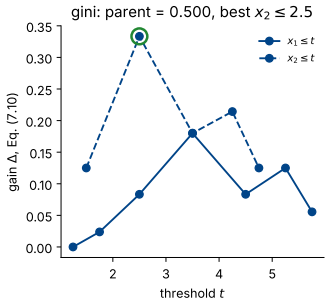
- A split that moves one point ($x_1 \leq 1.25$: points 1 and 9, one of each class) gains nothing: the children are as mixed as the parent.
- Symmetric splits give symmetric rows ($x_1 \leq 2.5$ and $x_1 \leq 4.5$).
- $x_2 \leq 2.5$ wins with both criteria, and by a margin: it creates a **pure** child of four points.
- The Gini index and the entropy *rank* the twelve splits identically here – the usual experience.

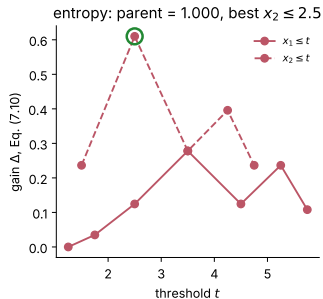
Then recurse

The left child is pure: a leaf (class 0). The right child has six points, $(1, 5)$: scan again, with its own midpoints.

The gain of every candidate, and the tree that results







Split 2, in the six points above the green line: $x_1 \le 2.75$ isolates points 4 and 9 (one of each class) from four class-1 points, $C = \frac{2}{6} \cdot 0.5 = 0.167$, $\Delta = 0.278 - 0.167 = 0.111$ – the best available.

Split 3, in the two remaining points: $x_1 \le 1.25$ (or $x_2 \le 4.5$ – a tie, broken by feature order) separates them. Depth 3, four leaves, training error zero. This is the tree of the vocabulary frame.

Pause and think 3: one row of the table by hand

Question

Verify the row $x_1 \leq 3.5$ of the table with the Gini index: which points go left, which right, what are G_{left}, G_{right}, C and Δ? Why does this split lose to $x_2 \leq 2.5$ although it also balances the classes nicely?

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Question

Verify the row $x_1 \leq 3.5$ of the table with the Gini index: which points go left, which right, what are G_{left}, G_{right}, C and Δ? Why does this split lose to $x_2 \leq 2.5$ although it also balances the classes nicely?

Answer

Left: points 1, 2, 3, 4 (class 0) and 9 (class 1), proportions $(4/5, 1/5)$, $G_{\text{left}} = 1 - 0.64 - 0.04 = 0.32$.
Right: point 5 (class 0) and 6, 7, 8, 10 (class 1), proportions $(1/5, 4/5)$, $G_{\text{right}} = 0.32$.
$C = \frac{5}{10}0.32 + \frac{5}{10}0.32 = 0.32$, $\Delta = 0.5 - 0.32 = 0.18$.

It leaves one stray point on *each* side, so both children must be split again; $x_2 \leq 2.5$ leaves one stray point on one side and none on the other. A pure child is worth more to a concave measure than two nearly-pure ones – the same lesson as the $(400, 400)$ example.

A worked example with categorical features: the ride data (Table 7.1)

Day	Outlook	Temperature	Humidity	Wind	Ride
1	Sunny	Hot	High	Weak	0
2	Sunny	Hot	High	Strong	1
3	Overcast	Hot	High	Weak	1
4	Rain	Mild	High	Weak	1
5	Rain	Cool	Normal	Weak	1
6	Rain	Cool	Normal	Strong	0
7	Overcast	Cool	Normal	Strong	1
8	Sunny	Mild	High	Weak	0
9	Sunny	Cool	Normal	Weak	1
10	Rain	Mild	Normal	Weak	1
11	Sunny	Mild	Normal	Strong	1
12	Overcast	Mild	High	Strong	1
13	Overcast	Hot	Normal	Weak	1
14	Rain	Mild	High	Strong	0

The question

From the weather, do we go for a ride? Four categorical features, a binary target, fourteen days.

How a categorical feature is split

The ID3 way: one child per value (outlook $\rightarrow$ sunny, overcast, rain) and the weighted impurity of Eq. (7.9) over all children. CART (and `scikit-learn`) instead makes binary splits on one-hot columns, "overcast or not?" – we shall see both.

What we compute

Gini index and entropy at the root, after a split on each feature, then the same in every impure branch until all leaves are pure.

The root, Eq. (7.13)

Ten rides, four non-rides

$$p_1 = \frac{10}{14}, \quad p_0 = \frac{4}{14}: \qquad G_{\text{root}} = 1 - \left(\tfrac{10}{14}\right)^2 - \left(\tfrac{4}{14}\right)^2 = 1 - 0.5102 - 0.0816 = 0.4082,$$

$$S_{\text{root}} = -\tfrac{10}{14}\log_2\tfrac{10}{14} - \tfrac{4}{14}\log_2\tfrac{4}{14} = 0.3467 + 0.5164 = 0.8631 \text{ bits.}$$

Splitting on Outlook: three children

value	days	(no, yes)	G	S
Sunny	1, 2, 8, 9, 11	(2, 3)	$1 - 0.16 - 0.36 = 0.48$	0.971
Overcast	3, 7, 12, 13	(0, 4)	0	0
Rain	4, 5, 6, 10, 14	(2, 3)	0.48	0.971

$$C_{\text{Gini}} = \tfrac{5}{14}0.48 + \tfrac{4}{14}0 + \tfrac{5}{14}0.48 = 0.3429, \qquad C_{\text{ent}} = \tfrac{10}{14}0.971 = 0.6935,$$

Gini gain $= 0.4082 - 0.3429 = 0.0653$, information gain $= 0.8631 - 0.6935 = 0.1696$.

All four features at the root (Table 7.2)

feature	Gini after	Gini gain	entropy after	info. gain
Outlook	**0.3429**	**0.0653**	**0.6935**	**0.1696**
Humidity	0.3673	0.0408	0.7885	0.0747
Temperature	0.4048	0.0034	0.8571	0.0060
Wind	0.4048	0.0034	0.8571	0.0060

Humidity, for the record: High = days 1, 2, 3, 4, 8, 12, 14, counts $(3, 4)$, $G = 24/49 = 0.4898$; Normal = days 5, 6, 7, 9, 10, 11, 13, counts $(1, 6)$, $G = 12/49 = 0.2449$; $C = \frac{7}{14}(0.4898 + 0.2449) = 0.3673$.

Both criteria choose Outlook

And for a visible reason: of its three values, **overcast is pure** – all four overcast days are rides – so that branch is a leaf at once. Humidity is next best; temperature and wind are nearly useless at the root.

This is the first step of ID3

Which attribute best separates the examples? Select it, make one branch per value, sort the examples into the branches, repeat in each impure branch. Greedy, never backtracking.

Down the tree: the two impure branches

Rain: days 4, 5, 6, 10, 14, counts $(2,3)$, $G = 0.48$

Wind: Weak = days 4, 5, 10, all rides; Strong = days 6, 14, no rides. Both children pure, $C = 0$, gain 0.48 (entropy: gain 0.971). Humidity gives $C = 0.4$, Temperature 0.467. **Wind** wins and the branch is finished.

Sunny: days 1, 2, 8, 9, 11, counts $(2,3)$, $G = 0.48$

Humidity: High = days 1, 2, 8, counts $(2,1)$, $G = 0.444$; Normal = days 9, 11, pure.
$C = \frac{3}{5}0.444 = 0.267$.
Wind: Weak = days 1, 8, 9, counts $(2,1)$; Strong = days 2, 11, pure. $C = 0.267$ – **a tie**.
Temperature: Hot $(1,1)$, Mild $(1,1)$, Cool pure:
$C = \frac{4}{5}0.5 = 0.4$.
Our code takes the first in feature order, Humidity; its High child $(2,1)$ is then split by Wind: Weak = days 1, 8 (no ride), Strong = day 2 (ride).

The finished tree

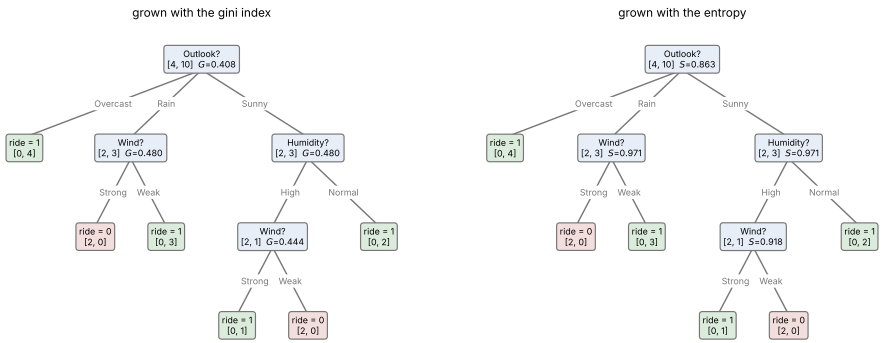
- Outlook = Overcast → ride.
- Outlook = Rain → Wind: Weak → ride, Strong → no.
- Outlook = Sunny → Humidity: Normal → ride; High → Wind: Strong → ride, Weak → no.

Depth 3, seven leaves, every leaf pure, training error zero.

Gini or entropy?

The two criteria rank every split the same way at every node here, so they grow the **same tree**; only the numbers on the nodes differ. The tie on the sunny branch is real for both.

The ride tree, grown by our code with both criteria



Node labels: [no rides, rides] and the node impurity. The code (`build_id3` in the notebook) is fifteen lines: compute the weighted impurity of Eq. (7.9) for every remaining feature, take the smallest, make one child per value, recurse.

The same data in scikit-learn

```
from sklearn.preprocessing import OneHotEncoder
from sklearn.tree import (DecisionTreeClassifier,
                          plot_tree, export_text)

X, y, features = ride_data() # 14 x 4 strings, 0/1
enc = OneHotEncoder(sparse_output=False)
Xoh = enc.fit_transform(X) # 14 x 10 indicators
names = [n.replace("_", "=")
         for n in enc.get_feature_names_out(features
              )]

clf = DecisionTreeClassifier(criterion="entropy")
clf.fit(Xoh, y)
print(export_text(clf, feature_names=names))
plot_tree(clf, feature_names=names,
          class_names=["no ride", "ride"], filled=
               True)
```

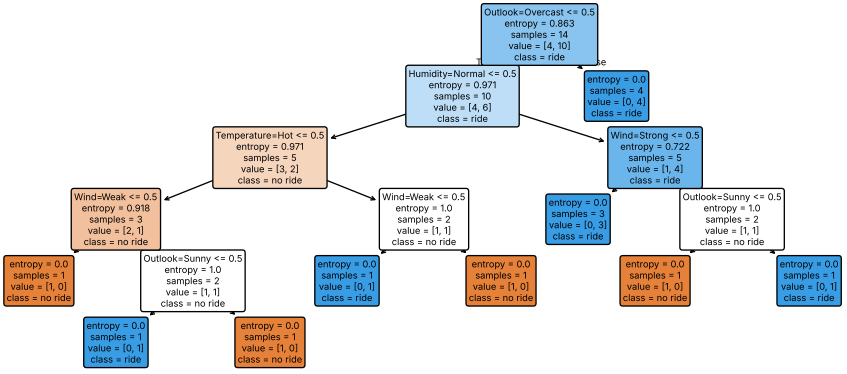
What is different

`scikit-learn` is CART: every split is binary, and a categorical feature must first be **one-hot encoded** – ten indicator columns "Outlook=Sunny", "Wind=Weak", ... The root asks "Outlook=Overcast ≤ 0.5?", i.e. *is it overcast?*, which is our overcast leaf; then "Humidity=Normal?", and so on.

What is the same

The impurity (`criterion="gini"` or `"entropy"`), the greedy search, pure leaves, training error zero. Binary splits need more levels: depth 5 and nine leaves against our depth 3 and seven.

plot_tree on the one-hot ride data



Each box: the test, the node entropy, the number of samples, the counts [no ride, ride] and the majority class. Colour saturation is purity.

A decision tree from scratch, 1: the impurities and a node (Section 7.6)

```python
import numpy as np

def gini(y, classes):
    """Eq. (7.7): G = 1 - sum_k p_k^2."""
    p = np.array([np.mean(y == c) for c in classes
        ])
    return 1.0 - np.sum(p**2)

def entropy(y, classes):
    """Eq. (7.8), with 0 log 0 = 0."""
    p = np.array([np.mean(y == c) for c in classes
        ])
    p = p[p > 0]
    return -np.sum(p * np.log2(p))

def mse_impurity(y):
    """Eq. (7.12)."""
    return np.mean((y - y.mean())**2) if len(y)
        else 0.0
```

```python
class Node:
    __slots__ = ("feature", "threshold",
                 "left", "right", "value", "n")

    def __init__(self, value=None, n=0):
        self.feature = self.threshold = None
        self.left = self.right = None
        self.value, self.n = value, n
```

The data structure

A node is either a **test** (`feature`, `threshold`, two children) or a **leaf** (`feature is None`); every node stores its prediction `value`, so any node can be a leaf if we stop there.

A decision tree from scratch, 2: the class

```
class DecisionTree:
    """CART for regression and classification.
    task: "classification" or "regression"; criterion: "gini" or "entropy";
    max_features: if set, only a random subset of features is tried at each split
                  -- this is what turns bagging into a random forest (Part 6)."""

    def __init__(self, task="classification", criterion="gini", max_depth=None,
                 min_samples_split=2, min_samples_leaf=1, max_features=None, rng=None):
        self.task, self.criterion = task, criterion
        self.max_depth = np.inf if max_depth is None else max_depth
        self.min_samples_split, self.min_samples_leaf = min_samples_split, min_samples_leaf
        self.max_features = max_features
        self.rng = np.random.default_rng() if rng is None else rng

    def _impurity(self, y):
        if self.task == "regression":
            return mse_impurity(y)
        return gini(y, self.classes_) if self.criterion == "gini" else entropy(y, self.classes_)

    def _leaf_value(self, y):
        if self.task == "regression":
            return y.mean() # the optimal constant
        counts = [np.sum(y == c) for c in self.classes_]
        return self.classes_[int(np.argmax(counts))] # majority vote
```

A decision tree from scratch, 3: the best split – Eqs. (7.9)–(7.10)

```
    def _best_split(self, X, y):
        n, p = X.shape
        features = np.arange(p)
        if self.max_features is not None and self.max_features < p:
            features = self.rng.choice(p, self.max_features, replace=False) # random forest
        parent = self._impurity(y)
        best_gain, best_j, best_t = 0.0, None, None
        for j in features:
            xs = X[:, j]
            uniq = np.unique(xs) # sorted distinct values
            if len(uniq) < 2:
                continue
            for t in (uniq[:-1] + uniq[1:]) / 2.0: # candidate midpoints
                mask = xs <= t
                nl, nr = mask.sum(), n - mask.sum()
                if nl < self.min_samples_leaf or nr < self.min_samples_leaf:
                    continue
                child = (nl / n) * self._impurity(y[mask]) + (nr / n) * self._impurity(y[~mask])
                gain = parent - child # Eq. (7.10)
                if gain > best_gain + 1e-12: # first maximum wins a tie
                    best_gain, best_j, best_t = gain, j, t
        return best_gain, best_j, best_t
```

This is the recipe frame, line for line: the double loop over features and midpoints, $C(k, t)$ as `child`, the gain, the argmax.

A decision tree from scratch, 4: grow, fit, predict

```
    def _build(self, X, y, depth):
        node = Node(self._leaf_value(y), len(y))
        if (depth >= self.max_depth or len(y) < self.
              min_samples_split
                or len(np.unique(y)) == 1): # stopping
                     rules
            return node
        gain, j, t = self._best_split(X, y)
        if j is None or gain <= 0:
            return node
        mask = X[:, j] <= t
        node.feature, node.threshold = j, t
        node.left = self._build(X[mask], y[mask], depth +
              1)
        node.right = self._build(X[~mask], y[~mask], depth
               + 1)
        return node

    def fit(self, X, y):
        X, y = np.asarray(X, dtype=float), np.asarray(y)
        if self.task == "classification":
            self.classes_ = np.unique(y)
        self.root_ = self._build(X, y, 0)
        return self
```

```
    def predict(self, X):
        X = np.asarray(X, dtype=float)
        out = []
        for x in X:
            node = self.root_
            while node.feature is not None:
                  # walk to a leaf
                node = (node.left if x[node.
                     feature] <= node.
                     threshold
                       else node.right)
            out.append(node.value)
        return np.array(out)
```

That is all

Recursion does the bookkeeping; the whole class is under sixty lines and handles regression, two impurities, depth and leaf-size limits, and the random feature subset we need in Part 6.

Checking it against `scikit-learn`: the moons

Moons data (week 39), $n = 400$, noise 0.3, 300 training and 100 test points, Gini index, both implementations:

	ours		`scikit-learn`	
depth	train	test	train	test
1	0.8267	0.7800	0.8267	0.7800
3	0.9000	0.8600	0.9000	0.8600
5	0.9633	0.8600	0.9633	0.8600
unlimited	1.0000	0.8600	1.0000	0.8400

Leaves: 2, 7, 18 and 33 in both. The whole unlimited tree takes 0.1 s in pure Python.

Identical where it should be

Same accuracies, same number of leaves at every depth. At unlimited depth the test accuracies differ by two points out of a hundred: **ties** between equally good splits, broken differently.

The last row is the lesson

The fully grown tree fits the training data perfectly and generalises *no better* than depth 3 – here slightly worse for `scikit-learn`. The variance term of the bias-variance decomposition, Eq. (2.52), growing without limit. Everything from Part 5 on is a way of controlling it.

Why a tree has high variance, 1: the leaf mean

Start from the decomposition of Eq. (2.52)

$\mathbb{E}\big[(y-\hat{f}(\mathrm{x}))^2\big] = \sigma^2 + \mathrm{Bias}^2\hat{f}(\mathrm{x}) + \mathrm{Var}\,\hat{f}(\mathrm{x})$, with σ^2 the noise variance. A regression tree predicts the mean of the n_m training responses in the leaf R_m that contains x:

$$\hat{f}(\mathrm{x}) = \frac{1}{n_m}\sum_{i\in R_m} y_i, \qquad \mathrm{Var}\big[\hat{f}(\mathrm{x})\,\big|\,\text{partition}\big] = \frac{\sigma^2}{n_m}.$$

Grown to completion: $n_m = 1$

The variance is σ^2 – the leaf repeats one noisy observation – and the bias is zero (the tree interpolates). Expected test error $\sigma^2 + 0 + \sigma^2 = 2\sigma^2$: zero training error, *twice* the noise on new data. The Runge experiment measured 0.0214 at full depth with $\sigma^2 = 0.01$.

Depth d: the trade-off in orders of magnitude

With $\approx n/2^d$ points per leaf, $\mathrm{Var} \approx \sigma^2\, 2^d/n$: it **doubles with every level**. For a smooth f on boxes of width $w \propto 2^{-d}$ the constant fit has $\mathrm{Bias}^2 \approx \frac{1}{12}f'^2 w^2 \propto 4^{-d}$: it quarters with every level. The sum has a minimum at a moderate depth – the U-shape of the test curve, 0.0181 at depth 4 for $n = 200$ – and the minimum is *far above* σ^2, because a staircase is a poor approximant of a smooth curve.

Why a tree has high variance, 2: the partition is random too

The conditional variance is only half the story

The cutpoints are *estimated* from the same noisy data. The empirical criterion (7.3), as a function of s, is flat near its minimum, so a small change in a few y_i moves the argmin – and because the search is greedy, a different root split changes *every* split below it. By the law of total variance,

$$\operatorname{Var} \hat{f}(x) = \underbrace{\mathbb{E}\big[\sigma^2 / n_{m(x)}\big]}_{\text{leaf mean}} + \underbrace{\operatorname{Var}\big[\mathbb{E}(\hat{f}(x) \mid \text{partition})\big]}_{\text{which box } x \text{ lands in}},$$

and the second term is large wherever x is near a cutpoint: the prediction jumps between two box means. This is the **instability** of Section 7.7.

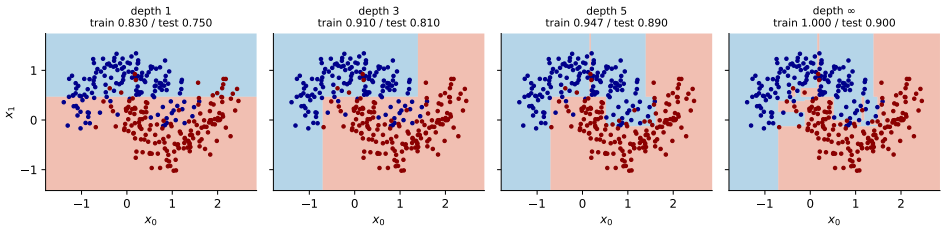
Compare a linear smoother

OLS, Ridge, kernel smoothers: $\hat{y} = Sy$, *linear* in y, so $\operatorname{Var} \hat{y}_i = \sigma^2 (SS^\top)_{ii}$ and on average $\sigma^2 \operatorname{tr}(S)/n = \sigma^2 p/n$ – bounded, continuous in the data, and small when $p \ll n$. A tree is a *discontinuous* function of its training data: no bound of this kind holds.

Degrees of freedom and optimism

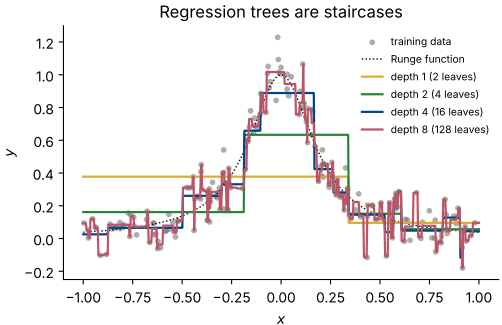
Conditional on the partition a tree with M leaves spends M degrees of freedom. Searching $p(n-1)$ candidate splits for the one that fits the noise best spends more: the *generalized* degrees of freedom $\sum_i \operatorname{Cov}(\hat{y}_i, y_i)/\sigma^2$ exceed M, and by Efron's identity the training error is optimistic by $2\sigma^2 \operatorname{df}/n$. Deep trees: low bias, huge df, high variance – precisely the profile that averaging repairs (Part 5).

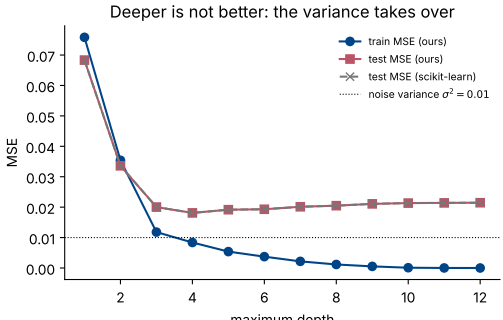
What the depth does to the boundary (book Figure 7.2)



At depth one a single threshold; at depth three the right gross shape; at depth five close to the truth; grown without limit, narrow rectangular islands around individual training points. Every boundary is a staircase of axis-aligned steps.

Regression trees are staircases, and deeper is not better





The Runge function with noise $\sigma = 0.1$, 200 training and 200 test points, our `DecisionTree("regression")`. A depth-d tree has at most 2^d leaves, each a constant. Train MSE falls to zero; test MSE is smallest at depth 4 (0.0181), then rises to 0.0214 at full depth – and never reaches the noise floor 0.01 that a degree-10 polynomial (0.0133) gets close to. A tree is the wrong tool for one smooth feature; `scikit-learn` gives the same test numbers to four digits.

Trees in `scikit-learn`: what you will actually call

```
from sklearn.tree import (DecisionTreeClassifier,
                          DecisionTreeRegressor, plot_tree)
from sklearn.model_selection import (train_test_split,
                                     GridSearchCV)

clf = DecisionTreeClassifier(criterion="gini", # or "entropy"
                             max_depth=5,
                             min_samples_leaf=5,
                             random_state=0)
clf.fit(X_train, y_train)
print(clf.score(X_test, y_test))
print(clf.get_depth(), clf.get_n_leaves())
print(clf.feature_importances_) # Gini decrease per feature
plot_tree(clf, filled=True, feature_names=names)

# choose the depth / leaf size / pruning by cross-validation
grid = GridSearchCV(DecisionTreeClassifier(random_state=0),
                    {"max_depth": [2, 3, 4, 6, 8, None],
                     "min_samples_leaf": [1, 5, 10],
                     "ccp_alpha": [0.0, 1e-3, 1e-2]},
                    cv=5).fit(X_train, y_train)
print(grid.best_params_, grid.best_score_)

reg = DecisionTreeRegressor(max_depth=4).fit(X_train, y_train)
```

The knobs, and what they are in our code

`criterion` = `gini`/`entropy` (`squared_error` for the regressor); `max_depth`, `min_samples_split`, `min_samples_leaf` = the three stopping rules of `_build`; `max_features` = the random subset in `_best_split`; `ccp_alpha` = the α of Eq. (7.4).

Two habits

No scaling pipeline is needed (it would change nothing). Cross-validate the depth or α exactly as you cross-validated λ in Project 1 and the learning rate in Project 2 – the week-44 machinery applies unchanged.

Strengths and weaknesses of a single tree (Section 7.7)

In favour

- Simple to understand, to interpret, to draw: a **white box** – any prediction is explained by its sequence of tests.
- Little data preparation: no scaling or centring; invariant under monotone transformations of the features.
- Prediction costs $\mathcal{O}(\text{depth})$.
- Numerical and categorical features, multiple outputs, missing values (with some care).

Against

- **Overfits** readily: depth limits, leaf sizes or pruning are mandatory.
- **Unstable**: a small change in the data can give a completely different tree – the high variance we just measured.
- Greedy: no guarantee of the best tree (NP-complete anyway).
- Staircases for diagonal boundaries; biased towards features with many split points and towards the majority class.

The crucial defect is the opportunity

A low-bias, high-variance, *unstable* predictor is exactly what averaging helps most. The rest of the lecture follows from that one observation.

Pause and think 4: scaling, logarithms and trees

Question

You standardise every feature (subtract the mean, divide by the standard deviation) before fitting (a) a decision tree, (b) Ridge regression, (c) a neural network trained with SGD. Which fits change? And if instead you replace a feature x_j by $\log x_j$ ($x_j > 0$)?

Pause and think 4: scaling, logarithms and trees

Question

You standardise every feature (subtract the mean, divide by the standard deviation) before fitting (a) a decision tree, (b) Ridge regression, (c) a neural network trained with SGD. Which fits change? And if instead you replace a feature x_j by $\log x_j$ ($x_j > 0$)?

Answer

(a) **Nothing** changes, in either case: a test $x_j \le t$ becomes $(x_j - \mu)/\sigma \le (t - \mu)/\sigma$, or $\log x_j \le \log t$ – the same points go left and right at every candidate split, so the same tree is grown with the same gains. Any strictly monotone map of a feature leaves a tree invariant.
(b) Ridge changes under both: the penalty $\lambda\|\boldsymbol{\theta}\|^2$ weighs coefficients whose meaning depends on the units, and $\log x_j$ is a different model. (c) The network's fit changes too (conditioning of the optimisation, and $\log x_j$ is a different input), which is why weeks 37–41 insisted on scaling. Trees are the one method in the course for which the advice "scale your data" is void.

Why averaging helps (Section 7.8): independent predictors, Eq. (7.14)

Average B predictors

$\hat{f}_1, \ldots, \hat{f}_B$, each unbiased for the target with variance σ^2, and $\bar{f} = \frac{1}{B}\sum_b \hat{f}_b$. If they are **independent**,

$$\mathbb{E}[\bar{f}] = \mathbb{E}[\hat{f}_b], \qquad \mathrm{var}(\bar{f}) = \frac{\sigma^2}{B}.$$

Read against the bias-variance decomposition, Eq. (2.52)

The bias is untouched; the variance falls as $1/B$. Averaging attacks the **variance term only**. That is why it is the right treatment for *deep* trees (bias already small, variance huge) and the wrong one for a model that is too rigid – averaging a hundred straight lines gives a straight line.

The catch

Predictors fitted to the same data set are *not* independent.

The central formula of the chapter, Eqs. (7.15)–(7.16)

Pairwise correlation ρ

$$\operatorname{var}(\bar{f}) = \frac{1}{B^2}\Big[\sum_b \operatorname{var}(\hat{f}_b) + \sum_{b\neq b'} \operatorname{cov}(\hat{f}_b, \hat{f}_{b'})\Big] = \frac{1}{B^2}\Big[B\sigma^2 + B(B-1)\rho\sigma^2\Big],$$

which simplifies to

$$\boxed{\operatorname{var}(\bar{f}) = \rho\,\sigma^2 + \frac{1-\rho}{B}\,\sigma^2.}$$

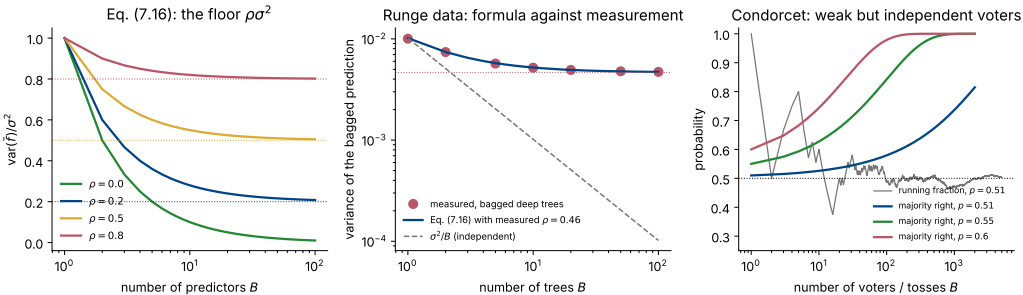
Read it carefully

The second term vanishes as $B \to \infty$: more predictors always help, with diminishing returns. The first term does **not depend on** B: however many trees we average, the variance cannot fall below $\rho\sigma^2$.

The design specification

Make σ^2 small (accurate members), B large (many members), and – easy to overlook – ρ small (**diverse** members). Bagging attacks σ^2/B; random forests exist because bagging leaves ρ too large; boosting (next week) attacks the bias instead.

Eq. (7.16) measured on bagged trees, and Condorcet's jury



Left: the formula; dotted lines are the floors $\rho\sigma^2$. **Middle**: 200 independent training sets of the noisy Runge function, 100 deep trees on bootstrap samples of each; measured $\sigma^2 = 0.0102$ for one tree, measured $\rho = 0.46$ between two trees of the same ensemble, and the variance of the average of B of them – the formula with these two numbers, no fitting. Averaging 100 trees removes half the variance and then stops. **Right**: B independent voters each right with probability p: a majority of 1001 voters with $p = 0.55$ is right with probability 0.999; with $p = 0.51$, 0.737. Weak but independent voters combine into a strong one – the italics are the assumption reality violates.

Bagging (Section 7.9): bootstrap aggregation

The recipe

We cannot draw B independent training sets – we have one. So we manufacture them by the **bootstrap** of week 36: draw n observations *with replacement*, fit a tree, repeat B times, and **average** the predictions (regression) or take a **majority vote** (classification).

Grow the trees deep and do not prune

Deliberate: deep trees have low bias and high variance; averaging removes variance and leaves bias. We want the members as unbiased as possible and let the ensemble handle the rest.

What it costs

B fits instead of one (embarrassingly parallel), B trees to store and walk at prediction time, and the readable tree is gone: a hundred trees cannot be drawn as one.

A free validation set: the out-of-bag error, Eq. (7.17)

A third of the data misses each tree

The probability that a given observation is *not* drawn in one bootstrap sample of size n is

$$\left(1 - \frac{1}{n}\right)^n \longrightarrow e^{-1} \approx 0.368 \qquad (n \to \infty).$$

Measured in our implementation: 0.3693 against 0.3679.

Out-of-bag prediction

For each observation, average (or let vote) only the trees that did **not** see it – about $0.37B$ of them. The resulting error is an almost unbiased estimate of the test error, with no separate split and no cross-validation loop. In `scikit-learn`: `oob_score=True`.

What survives of interpretability

Variable importance: the total decrease of impurity (MSE or Gini) due to splits on each feature, summed over all trees and normalised. A summary, not an explanation – and known to favour features with many distinct values.

Bagging from scratch, 1: fit and predict

```python
class BaggingClassifier:
    """Bootstrap aggregation of unpruned trees, with out-of-bag scoring."""

    def __init__(self, n_estimators=100, max_depth=None, max_features=None, rng=None):
        self.n_estimators, self.max_depth = n_estimators, max_depth
        self.max_features = max_features
        self.rng = np.random.default_rng(0) if rng is None else rng

    def fit(self, X, y):
        X, y = np.asarray(X, dtype=float), np.asarray(y)
        n = len(y)
        self.classes_ = np.unique(y)
        self.trees_, self.oob_ = [], []
        for _ in range(self.n_estimators):
            idx = self.rng.integers(0, n, n) # bootstrap resample, with replacement
            self.oob_.append(np.setdiff1d(np.arange(n), idx)) # points this tree never sees
            tree = DecisionTree("classification", "gini", max_depth=self.max_depth,
                                max_features=self.max_features, rng=self.rng)
            self.trees_.append(tree.fit(X[idx], y[idx]))
        return self

    def predict(self, X):
        X = np.asarray(X, dtype=float)
        votes = np.stack([t.predict(X) for t in self.trees_]) # (B, n_points)
        out = np.empty(X.shape[0], dtype=self.classes_.dtype)
        for i in range(X.shape[0]): # majority vote
            counts = [np.sum(votes[:, i] == c) for c in self.classes_]
            out[i] = self.classes_[int(np.argmax(counts))]
        return out
```

Bagging from scratch, 2: the out-of-bag score

```
def oob_score(self, X, y):
    """Eq. (7.17): each point is scored only by
    the trees that never saw it."""
    X, y = np.asarray(X, dtype=float), np.asarray(y)
    tally = np.zeros((len(y), len(self.classes_)))
    for tree, oob in zip(self.trees_, self.oob_):
        if len(oob) == 0:
            continue
        pred = tree.predict(X[oob])
        for k, c in enumerate(self.classes_):
            tally[oob[pred == c], k] += 1
    seen = tally.sum(axis=1) > 0
    vote = self.classes_[np.argmax(tally[seen], axis
        =1)]
    return np.mean(vote == y[seen])
```

Reading it

`tally[i, k]` counts how many of the trees that did not train on point i voted for class k. Each point gets about $0.37B$ votes; with $B = 100$ every point is seen in practice. The majority of those votes is the out-of-bag prediction.

Why it is honest

No tree that votes on point i has ever seen y_i: exactly the situation of a test point, obtained from the training set alone.

Bagging in scikit-learn

```
from sklearn.ensemble import (BaggingClassifier,
                              BaggingRegressor)
from sklearn.tree import DecisionTreeClassifier

bag = BaggingClassifier(
       estimator=DecisionTreeClassifier(), # deep,
            unpruned
       n_estimators=100,
       max_samples=1.0, # bootstrap sample of size n
       bootstrap=True, # with replacement
       oob_score=True, # Eq. (7.17)
       n_jobs=-1, # trees are independent: parallel
       random_state=0)
bag.fit(X_train, y_train)
print(bag.oob_score_, bag.score(X_test, y_test))
print(len(bag.estimators_)) # the 100 trees
print(bag.predict_proba(X_test)[:3]) # vote fractions

# regression: averages instead of votes
breg = BaggingRegressor(n_estimators=200, oob_score=
     True)
```

Line by line

`estimator` is the base learner – any estimator works, trees are the standard choice; `max_samples=1.0` and `bootstrap=True` give the bootstrap of Eq. (7.17); `oob_score=True` makes the free validation estimate available as `oob_score_`; `predict_proba` returns the fraction of trees voting for each class.

The one parameter

$B =$ `n_estimators`. By Eq. (7.16) more trees never hurt, they only stop helping; choose by budget, and watch `oob_score_` flatten.

Random forests (Section 7.10): one tweak that decorrelates the trees

As bagging, plus

Each time a split is considered, only a **random sample of m of the p features** is offered as candidates, and the split must use one of them. A fresh sample at every split. Typically

$$m \approx \sqrt{p} \quad \text{(classification)}, \qquad m \approx p/3 \quad \text{(regression)}. \tag{7.18}$$

At each split the algorithm is *not allowed* to look at most of the features.

In our code it is one line

The `max_features` branch of `_best_split`: `features = rng.choice(p, max_features, replace=False)`.

Why this clever restriction works

The problem with bagging

Suppose one feature is very strong and several others moderately strong. Almost every bagged tree puts the strong feature at its root; the trees look alike and their predictions are **highly correlated**. By Eq. (7.16) the floor $\rho\sigma^2$ is high and more trees do not help.

The cure

With m of p features per split, the strong feature is unavailable in a fraction $1 - m/p$ of the splits; other features get their turn, and the trees *genuinely differ*: ρ falls.

The price

Each tree is worse than it could be: σ^2 rises. Random forests work because, for deep trees, the fall in ρ outweighs the rise in σ^2 – and m is the hyperparameter trading them off.

The two limits

$m = p$ is bagging exactly. $m = 1$ gives maximally decorrelated but very weak trees (the split *feature* is random, only the threshold is chosen). Eq. (7.18) is a rule of thumb; m deserves cross-validation like any other hyperparameter.

A random forest from scratch: a subclass

```
class RandomForestClassifier(BaggingClassifier):
    """Bagging plus a random subset of
    m features at every split."""

    def __init__(self, n_estimators=100, max_depth=
          None,
                 max_features="sqrt", rng=None):
        super().__init__(n_estimators, max_depth,
             None, rng)
        self._mf = max_features

    def fit(self, X, y):
        p = np.asarray(X).shape[1]
        self.max_features = (int(np.sqrt(p))
                             if self._mf == "sqrt"
                                 else self._mf)
        return super().fit(X, y) # Eq. (7.18)
```

Where the randomness enters

Twice: the bootstrap sample of each tree (inherited from `BaggingClassifier.fit`), and the feature subset drawn *inside every call* of `_best_split`, because `max_features` is passed down to each `DecisionTree`.

A detail

On the moons $p = 2$, so $m = \lfloor \sqrt{2} \rfloor = 1$: every split picks its feature at random. On the cancer data below, $p = 30$ and $m = 5$.

Single tree, bagging, forest on the moons

$n = 500$, noise 0.3, 375 training and 125 test points, $B = 100$ deep trees:

	test accuracy	out-of-bag
single tree (ours)	0.824	–
single tree (`sklearn`)	0.824	–
bagging (ours)	0.896	0.907
bagging (`sklearn`)	0.872	0.923
forest (ours)	0.880	0.909
forest (`sklearn`)	0.880	0.915

Out-of-bag fraction measured: 0.3693 ($1/e = 0.3679$).

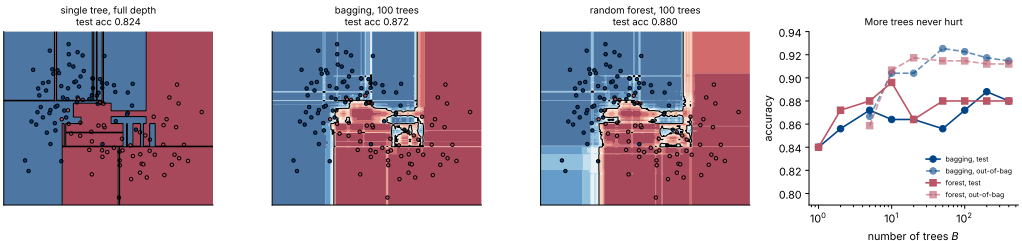
What is significant and what is not

Both ensembles beat the single tree by five to seven points. Between the four ensemble numbers the differences are *not* significant: for an accuracy near 0.9 on 125 points the standard error, Eq. (2.24), is $\sqrt{0.9 \cdot 0.1/125} \approx 0.027$.

Trust the out-of-bag numbers more

They use all 375 training points. They agree: around 0.91–0.92 for every ensemble, against 0.82 for one tree.

What the ensembles do to the boundary



Colour = fraction of trees voting for class 1 (`predict_proba`); black line = the majority. The single tree's islands around individual points are averaged away; the ensemble boundary is smoother and comes with a graded confidence. **Right**: test and out-of-bag accuracy against B – the gain is complete by $B \approx 50$, exactly the saturation of the second term of Eq. (7.16); nothing is lost by going further.

Random forests in `scikit-learn`

```
from sklearn.ensemble import (RandomForestClassifier,
                              RandomForestRegressor)

rf = RandomForestClassifier(
       n_estimators=200, # B: as many as you can afford
       max_features="sqrt", # m = sqrt(p), Eq. (7.18);
                            # also "log2", an int, a float, None (= bagging)
       max_depth=None, # grow deep
       min_samples_leaf=1,
       oob_score=True,
       n_jobs=-1,
       random_state=0)
rf.fit(X_train, y_train)
print(rf.oob_score_, rf.score(X_test, y_test))
imp = rf.feature_importances_ # mean Gini decrease
for i in np.argsort(imp)[::-1][:5]:
    print(feature_names[i], round(imp[i], 3))

# regression: max_features defaults to 1.0 (= p, i.e. bagging);
# set max_features=1/3 for the classical rule
rfr = RandomForestRegressor(n_estimators=200, max_features=1/3,
                            oob_score=True).fit(X_train, y_train)
```

Two hyperparameters that matter

`n_estimators` (B) and `max_features` (m). Everything else is the tree's own knobs, and a forest is remarkably insensitive to them. `max_features=None` turns the forest back into bagging.

Note the default for regression

Since version 1.1 `RandomForestRegressor` uses *all* features by default – bagging – so set `max_features` yourself if you want the decorrelation.

A real data set: the Wisconsin breast-cancer data

569 tumours, 30 features (radius, texture, concavity, ...), benign or malignant; 398 training and 171 test points; $B = 200$:

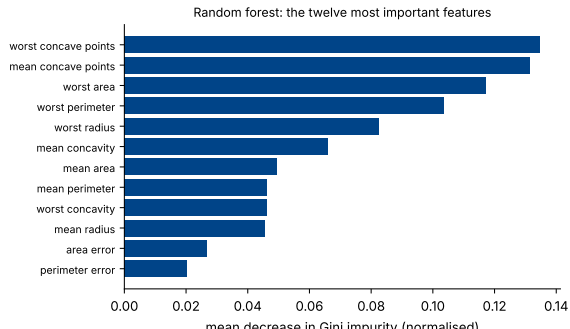
	train	test	out-of-bag
tree, depth 3	0.977	0.918	–
tree, full depth (7)	1.000	0.930	–
bagging	1.000	0.942	0.955
random forest, $m = 5$	1.000	0.953	0.957
logistic regression, scaled	0.993	0.965	–

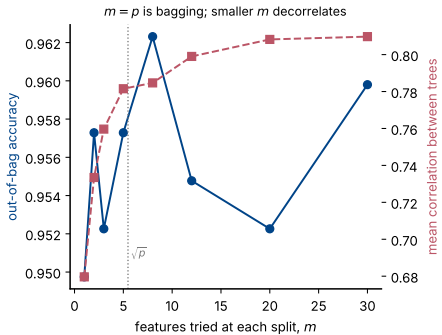
Mean pairwise correlation between the trees' test predictions: **bagging** 0.81, **forest** 0.78.

Three lessons

1. The ensembles beat the single tree again, by two to three points (standard error here ≈ 0.017).
2. The forest decorrelates only a little on these data: the 30 features are themselves strongly correlated (radius, perimeter, area), so a strong one is rarely *all* unavailable.
3. **Trees are not always best.** A scaled logistic regression wins on this nearly linearly separable problem. Forests are the safe default on tabular data, not the guaranteed winner.

Feature importances, and m as a dial for ρ





Left: the forest's variable importances – concave points, area, perimeter, radius; what replaces the readable tree. **Right**: the mean correlation between trees rises monotonically with m, from 0.68 at $m = 1$ to 0.81 at $m = p = 30$ (bagging), as Eq. (7.16) predicts; the out-of-bag accuracy moves within ± 0.005 – four tumours – so on these data any m from 2 to 30 is as good as any other. The dial works; here it has little to turn.

Pause and think 5: a forest with one feature

Question

We fit `RandomForestRegressor` and `BaggingRegressor`, 200 deep trees each, to the noisy Runge data of the regression frame ($p = 1$). Test MSE: single tree at depth 4: 0.0181, full depth: 0.0214; bagging: 0.0173; forest: 0.0173; degree-10 polynomial: 0.0133. Why are bagging and the forest identical here, why do they beat the single tree only modestly, and why does the polynomial still win?

Pause and think 5: a forest with one feature

Question

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Answer

With $p = 1$ there is nothing to subsample: $m = 1 = p$, and a forest *is* bagging – the decorrelation trick needs several features. The ensemble removes the variance of the full-depth tree ($0.0214 \to 0.0173$) but not its bias: an average of staircases is still a staircase, so the smooth peak and the flat tails are approximated with the same resolution limit – and a tree cannot extrapolate past the outermost training points. A polynomial is the right inductive bias for one smooth function; trees and forests earn their keep when there are *many* features, interactions, mixed types and no good parametric form – which is most tabular data.

Random forests as the off-the-shelf method for tabular data

Why people reach for them first

- No feature scaling, insensitive to monotone transformations, mixed data types, tolerant of missing values and outliers.
- A free error estimate: out-of-bag, Eq. (7.17).
- Rarely overfit as B grows: more trees only shrink the second term of Eq. (7.16), never increase the bias.
- Essentially two hyperparameters, one of which is "as many trees as you can afford".
- Embarrassingly parallel to train and to predict.

Their weaknesses follow from the same structure

- **No extrapolation**: every prediction is an average of observed targets.
- Axis-aligned boundaries, staircases for anything diagonal.
- Large to store, slow to evaluate compared with a linear model; the readable tree is gone.
- They attack variance only. When the single tree is *biased*, averaging cannot help – which is where **boosting** comes in.

Tomorrow, Project 3, and next week

Tomorrow: flipped classroom

About 20 minutes of questions and answers on today's material – bring the things you did not follow (the gain formula, the tie-breaking, the out-of-bag idea, Eq. (7.16)). Then we go through **Project 3** together.

Project 2

The deadline is **today, Monday November 9**.

Reading

Chapter 7, Sections 7.1–7.10 (7.6 and 7.10 contain the code of today); Hastie et al. 9.2 and 15; the code is in `week46.ipynb`.

Next Monday, week 47: boosting

Many *weak* learners in sequence, each correcting its predecessors – attacking the bias that averaging leaves alone: AdaBoost and the exponential loss, gradient boosting as gradient descent in function space, XGBoost, and boosting any loss with automatic differentiation (Sections 7.11–7.15). The gated recurrent cells (LSTM, GRU) get their own lecture later.

Summary of week 46

A single tree

Boxes and constants, grown top-down and greedily. Splits are scored by an impurity: MSE for regression, Gini index or entropy for classification – concave measures, so a pure child is rewarded. CART minimises the weighted child impurity $C(k, t)$ of Eq. (7.9), i.e. maximises the gain (7.10), over every feature and every midpoint; the ride data and ten points in the plane showed the arithmetic, and sixty lines of Python reproduce `scikit-learn`. A full-depth tree has zero training error and large variance.

Many trees

$\operatorname{var}(\bar{f}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$ (7.16). Bagging: bootstrap samples, deep trees, average or vote, out-of-bag error for free (7.17); it shrinks the second term. Random forests: $m \approx \sqrt{p}$ random features per split (7.18); it lowers ρ, hence the floor. Both beat a single tree by a wide margin and are hard to overfit with B; neither can extrapolate or reduce bias.

Next

Boosting: the route that attacks the bias.